A Real-Time Grasping Detection Network Architecture for Various Grasping Scenarios

Hejia Gao, *Member, IEEE*, Junjie Zhao, Juqi Hu, *Member, IEEE*, and Changyin Sun, *Senior Member, IEEE*

***Abstract*— In the field of robot grasping detection, due to uncertain factors such as different shapes, distinct colors, diverse materials, and various poses, robot grasping has become very challenging. This article introduces a integrated robotic system designed to address the challenge of grasping numerous unknown objects within a scene from a set of α-channel images. We propose a lightweight and object-independent pixel-level generative adaptive residual depthwise separable convolutional neural network (GARDSCN) with an inference speed of around 28 ms, which can be applied to real-time grasping detection. It can effectively deal with the grasping detection of unknown objects with different shapes and poses in various scenes and overcome the limitations of current robot grasping technology. The proposed network achieves 98.88% grasp detection accuracy on the Cornell dataset and 95.23% on the Jacquard dataset. To further verify the validity, the grasping experiment is conducted on a physical robot Kinova Gen2, and the grasp success rate is 96.67% in the single-object scene and 94.10% in the multiobject cluttered scene.**

***Index Terms*— Convolutional neural network, grasping detection, integrated robotic system, lightweight and object-independent, numerous unknown objects.**

I. Introduction

A. Background

WITH the rapid advancement of artificial intelligence, robot grasping technology has achieved certain breakthroughs. Presently, this technology has penetrated into multiple fields such as modern industrial, medical, and service industries, serving as a crucial force in driving social progress [1]. Current grasping scenarios are typically confined to structured environments such as assembly lines, where robots perform fixed and repetitive actions through preprogramming [2]. The types of objects targeted are often limited, and their placement patterns are usually consistent. To enable robots to grasp objects in diverse environments and perform better in complex scenarios, they need the capability to process environmental information and make intelligent decisions, adapting to increasingly complex and diverse task scenarios. Hence, robotic grasping remains a challenging open problem.

B. Related Works

The purpose of robot grasping detection is to find a set of grasping configurations (including grasping point position, joint posture, etc.) for a given scene containing objects. Under this configuration, the robot can stably grasp the corresponding objects by closing its fingers [3], [4]. The primary methodologies can be categorized into two major classes: analytic method and data-driven method.

1) Analytic Method: Analytic method refers to the method of calculating the grasp pose that meets the requirements through the kinematics and dynamics relationship according to the geometric information and physical model of the object and the robot [5]. First, numerous grasp poses are generated based on the known object model and saved as a database, which is done offline. Upon the appearance of an object within the scene, the robot performs object pose detection, transforms the grasp poses corresponding to that object from a database, and subsequently refines the final grasp pose through collision detection and filtering processes [6].

2) Data-Driven Method: Data-driven method focuses on how to extract abstract features from perceptual information and finally map them to the score of each grasp pose in the grasp space [7], thus reducing or completely eliminating the work of manual modeling and providing robots with better perception and adaptability. The data-driven method first involves the construction of a knowledge repository, during the offline phase, which maps perceptual information (such as RGBD images, point clouds, and tactile data) to grasp poses [8]. Subsequently, this knowledge repository is used to train deep neural networks, enabling the network to extract features from input perceptual information and map these features to a score for each grasp pose within the grasp space [9], [10].

In recent years, deep learning technology has been widely used in the field of robot grasping detection and has gradually become the dominant technology of data-driven method. Numerous researchers both domestically and internationally have conducted extensive investigations within the field of robot grasping detection [11], [12]. Fang et al. [13] proposed a unified spatio-temporal approach called Anygrasp.

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Hejia Gao and Junjie Zhao are with the School of Artificial Intelligence, Anhui University, Anhui 230601, China, also with the Anhui Provincial Key Laboratory of Security Artificial Intelligence, Anhui University, Anhui 230601, China, and also with the Engineering Research Center of Autonomous Unmanned System Technology, Ministry of Education, Hefei, Anhui 230601, China (e-mail: cysun@seu.edu.cn).

Juqi Hu is with the School of Artificial Intelligence, Anhui University, Anhui 230601, China, and also with the Key Laboratory of Knowledge Automation for Industrial Processes of Ministry of Education, Anhui University, Anhui 230601, China (e-mail: jqhu@ahu.edu.cn).

Changyin Sun is with the School of Artificial Intelligence, Anhui University, Anhui 230601, China, also with the Anhui Provincial Key Laboratory of Security Artificial Intelligence, Anhui University, Anhui 230601, China, also with the Engineering Research Center of Autonomous Unmanned System Technology, Ministry of Education, Hefei, Anhui 230601, China, and also with the School of Automation, Southeast University, Nanjing 210096, China.

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The geometric processing module estimates dense 7-degree-of freedom (DoF) grasp configurations from monocular sensory observations in a feedforward channel, while the temporal correlation module identifies grasp correspondences between these large grasp poses across every two observations. Ren et al. [14] proposed a pixel-level grasp detection network called TDDM-Net, which incorporates dual deconvolution and multidimensional attention. The dual deconvolution network is used to extract features and refine the grasp region. Furthermore, the introduction of the multidimensional attention mechanism enables the model to accurately identify the target object within the grasp region and determine the optimal grasp pose. Ainetter and Fraundorfer [15] introduced an end-to-end deep neural network grasp pose detection model that combines pixel-level semantic segmentation. This network comprises two branches: one dedicated to grasp pose detection and the other to semantic labeling of objects in the input image. Finally, the prediction information from both the branches is fused and fed into a multilayer perceptron for refinement, resulting in more accurate grasp outcomes. Wang et al. [16] proposed the AFFGA-Net to predict pixel-level OAR models on RGB images. By leveraging a hybrid atrous spatial pyramid to enhance feature utilization and increase effective weights, the adaptive decoder provides features of different scales for various grasp parameter detection tasks, ultimately outputting high-precision pixel-level grasp poses. Kim et al. [17] proposed DSQNet, which uses a deformable model as a representation of the object's geometry. This model is continuously updated to adapt to the specific characteristics of the encountered object. A supervised deep learning network is used to recognize and grasp the target object. DSQNet demonstrates excellent performance in grasping unknown and partially occluded objects.

Overall, the application of deep learning in the field of robotic grasp detection is a prevailing trend. Grasp detection is evolving toward finer granularity, ranging from pregenerating grasp candidates for subsequent evaluation and classification, to directly generating directed grasp boxes, and ultimately predicting grasp confidence and poses for each pixel. However, these methods are evaluated in idealized and single scenarios without clutter, resulting in models with limited generalization capabilities that greatly constrain their practical applications in real-world scenarios. In diverse and complex working environments, the success rate of robotic grasping significantly decreases. Based on the above, in this article, a lightweight grasping detection network generative adaptive residual depthwise separable convolutional neural network (GARDSCN) is introduced to address the prediction of grasp poses for novel multiple objects in diverse and complex scenarios, which balances grasp accuracy and detection speed.

Unlike previous research in robotic grasping, where the anticipated grasp involves a grasp rectangle determined through the selection of the optimal grasp from multiple grasp probabilities. Our network generates grasp confidence image, grasp width image, and grasp angle image, from which we can not only infer the optimal grasp of a single object but also predict multiple grasps of a single object and multiple grasps of multiple objects. Furthermore, our designed robot grasping system integrates the ROS interface, so it can be applied to any robot, making up for the poor platform portability of the existing advanced methods. Table I presents a comparative analysis of our study against recent research in the field of robot grasping for unknown objects.

TABLE I
COMPARISON OF OUR WORK WITH EXISTING RELATED WORKS

	[18]	[19]	[20]	[21]	[22]	[23]	[24]	**Ours**
Real robot experiments	✓	✓	✓	✓	✓	✓	✓	✓
Results on Cornell	✓	✓	✓	✓	✓	✓	✗	✓
Results on Jacquard	✗	✗	✗	✗	✓	✓	✗	✓
Multi-object cluttered scene	✗	✓	✗	✓	✓	✓	✓	✓
Transparent objects	✗	✗	✗	✗	✗	✗	✗	✓
Single-object multiple-grasp	✓	✓	✗	✓	✓	✓	✗	✓
Multi-colored background	✗	✗	✗	✗	✗	✗	✗	✓
Varying illumination	✗	✗	✗	✗	✗	✗	✗	✓

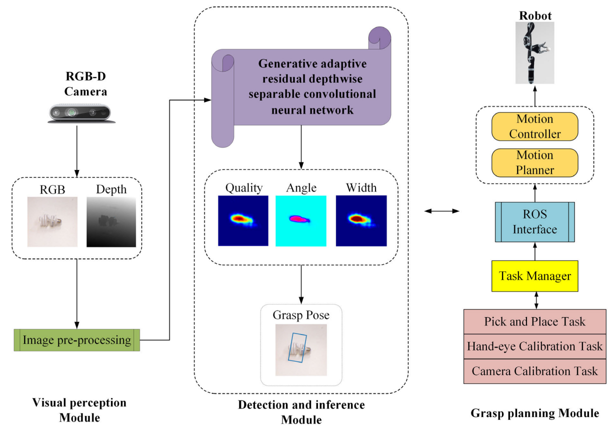

Fig. 1. Architecture of the intelligent robot grasping system.

C. Motivation and Contributions

Most existing studies have designed deep-learning-based methods to address robot grasping in unstructured task scenarios [25], [26], [27]. However, a significant amount of this work focuses on designing and testing networks under ideal constrained conditions, which are often impractical in many real-world situations. Real-world environments are typically more complex, incorporating factors such as background interference, lighting variations, and a wide range of objects, which severely limit the application of designed models in areas such as warehousing, logistics, and home services. To address the challenges of robot grasping in diverse and complex scenarios, this article proposes a novel systematic robot grasping system and an efficient and robust GARDSCN.

Fig. 1 illustrates the architecture of our novel robot grasping system, which primarily consists of three components: visual perception module, detection and inference module, and grasping planning module. The visual perception module collects scene visual information and performs image registration. The detection and inference module includes a pretrained grasping detection network and grasping parameter decoding, which decodes the network output into a grasping pose representation. The grasping planning module involves the transformation of the grasping representation model, followed by the calculation of the robot's motion path based on inverse kinematics and the control of the robot's grasping. As our system integrates ROS interfaces, it can be mounted on any robot. We also propose the GARDSCN grasping detection network that can generate pixel-level symmetric grasping for various unknown objects in diverse and complex scenarios. Compared with the existing methods, our network has a significantly reduced number of parameters, resulting in extremely fast inference speed of only 28 ms, while maintaining a high grasping success rate.

The main contributions of this article are as follows.

1) By integrating visual collection, grasping detection, and motion control, a novel robot grasping system is designed and this grasping system can be easily applied to various robot platforms undertaking different tasks. In real-world grasping tasks, our system can achieve success rates of 96.67% in single-object scene and 94.10% in complex multiobject cluttered scene.

2) The designed visual attention residual multidimensional convolution (VARMC) in the proposed robot grasping system is adaptively focusing on crucial channel and spatial feature information. It can adaptively adjust the weights of different features, enabling it to focus more on the crucial features for grasping tasks while suppressing the interference of background information.

3) The novel GARDSCN in the proposed robot grasping system can achieve the robust pose extrapolating for complex and various scenarios even under the change in illumination and background. GARDSCN exhibits an inference speed of 28 ms and achieves grasping accuracy of 98.88% and 95.23% on the publicly available Cornell and Jacquard datasets, respectively. Compared with the existing advanced methods, the number of parameters and the training time are reduced by near eight times and six times, respectively.

II. Problem Description

A. How Can We Properly Represent a Grasp?

In this section, in the face of numerous novel objects with different shapes, a general grasping representation model is necessary and important [17], [28].

We consider the 2-D grasping detection problem: given the input RGB or Depth image, and infer the optimal grasp pose in the scene. The existing studies have used a 5-D grasp representation: a directed grasp rectangular box to represent the grasp pose [7], [29]

$$g = (x, y, w, h, \theta) \tag{1}$$

where $(x,\ y)$ represents the center point of the grasp rectangular bounding box in the image coordinate system; w signifies the width of the rectangular box, which represents the projected width of the gripper's opening in the image; h corresponds to the height of the rectangular box; and θ denotes the angle formed by the rectangular box with respect to the x-axis of the image coordinate system. For the grasping detection task, this form has great limitations [30], [31]. To adapt to the proposed network, for a given image, we define a new grasp pose representation in the image coordinate system

$$G_I = (G_k, \theta_k, W_k, Q) \tag{2}$$

where $G_k = (m, n)$ denotes the pixel grasp point in the image coordinate system; θ_k is the rotation angle of the end gripper with respect to the image's horizontal axis, which is represented in [$-\pi/2$, $\pi/2$]; and W_k represents the pixel width of the gripper's opening in the image coordinate system. Due to the physical properties of the end gripper of robot itself, W_k is expressed as the pixel width in the range of [0, $W_{\max}$], and $W_{\max}$ is the pixel maximum width that the end gripper can open. Q is the grasp confidence (grasp quality score), and its value is in [0, 1]. The value closer to 1 indicates a higher grasp success rate, while the value closer to 0 signifies a lower grasp success rate.

However, when the physical robot is grasping, the robot needs to receive 3-D grasp pose data to complete the complex grasping task. Therefore, the grasp pose in the robot coordinate system of 3-D space can be described as

$$G_m = (S, \Phi_m, W_m, Q) \tag{3}$$

where $S = (x, y, z)$ is the center position of the end gripper in the robot coordinate system. Φ_m is the angle of rotation of the gripper with respect to the camera frame. W_m is the actual width of the gripper opening, and Q is the same scalar as (2).

To execute the identified grasp pose within the image coordinate system on the physical robot, a conversion from the 2-D grasp pose in the image coordinate system to a 3-D grasp pose in the robot coordinate system is required. Assuming knowledge of the hand–eye calibration results and camera intrinsic parameters, this transformation can be performed as follows:

$$G_m = T_{\mathrm{rc}}(T_{\mathrm{ci}}(G_I)) \tag{4}$$

where T_{ci} denotes the intrinsic parameters of the camera, which establish the transformation relationship between the pixel coordinate system and the camera coordinate system, enabling the conversion of the grasp pose from two dimensions to three dimensions. T_{rc}, obtained from the hand–eye calibration task, is used to describe the rotational relationship between the camera coordinate system and the robot's end-effector coordinate system.

The grasping set in the image space is represented as follows:

$$G = (\Theta, W, Q) \in \mathbb{R}^{3\times h\times w} \tag{5}$$

where Θ, W, and Q are the grasping sets of the grasp angle Θ_k, the grasp width W_k, and the grasp confidence Q calculated at each pixel of the image using (2), respectively. The representation of the grasping set ensures that the subsequent selection of a set of grasping configurations with the highest confidence Q is guaranteed, $(G_m)^* = \max_Q G$.

III. Approach

We propose an integrated intelligent robot grasping system: visual perception module, detection and inference module, and grasp planning module. The proposed system architecture is shown in Fig. 1. The visual perception module collects the information of the target objects from the RGBD camera. In the detection and inference module, a novel network GARDSCN is proposed to calculate the grasp pose of objects. After obtaining the optimal grasp pose calculated by the detection and inference module, the grasp planning module plans a safe motion trajectory, which avoids collision with the external environment or objects, and finally allows the grasped objects to be safely transported to the target position.

A. Visual Perception Module

The module consists of a machine vision unit: the Realsense D435i camera, which is fixed at the end of the robot arm to collect visual information of the external scene. The visual information includes RGB images and the corresponding Depth images. After image completion, denoising, cropping, normalization, and other operations, the preprocessed RGB image and Depth image information are output.

B. Detection and Inference Module

The module receives preprocessed RGB images and Depth images from the visual perception module. The proposed GARDSCN in the module extracts features from these image data to produce grasp confidence maps, grasp angle maps, and grasp width maps for the target objects. Our detection and inference module can output the highest quality grasp for a single object, multiple grasps for a single object, and multiple grasps for multiple objects.

C. Grasp Planning Module

The module is composed of a motion planner and a motion controller. The module receives grasp pose calculated from the detection and inference module and converts the grasp pose into a 3-D grasp pose in the robot coordinate system by combining the results obtained from the hand–eye calibration task. Furthermore, using the ROS interface, a motion planner is used to generate a collision-free and secure trajectory. Subsequently, a motion controller controls the robot to the target position, facilitating the execution of grasping and placement tasks. Simultaneously, the motion state of robot is monitored in real-time.

D. VARMC Module

Recently, although convolutional neural networks have been extensively used in robot grasping detection, the existing methods assign the same weight to all the features on all the positions or image channels when extracting features from images, which cannot suppress features in nongrasping areas, resulting in the network unable to obtain important feature information. Moreover, the traditional convolution parameters are large and the operation cost is high, which leads to the reduction of real-time performance. More importantly, as the network deepens, not only does the overall computational workload and processing time significantly increase but also the substantial growth in the number of parameters to be learned can lead to overfitting phenomena in the predictions of network. Therefore, to deepen the network while avoiding the above problems, this article proposes the VARMC module to solve these problems.

In this article, the VARMC module is shown in Fig. 2. The VARMC module initially uses a 3 × 3 depthwise convolution and a 1 × 1 convolution for feature extraction, allowing the network to autonomously select superior features. Depthwise convolution captures spatial features within each input channel but does not capture interchannel relationships. The role of the 1 × 1 convolution is to complement the missing functionality of the former, using 1 × 1 convolution kernels to modify the channel count without altering the feature map dimensions, effectively integrating channel information. ReLu activation functions and batch normalization are subsequently introduced after depthwise convolution and 1 × 1 convolution layers to increase the nonlinearity of VARMC, thereby expediting the training and convergence rates of the entire network. This design not only reduces the parameter number of VARMC but also lowers the computational costs and deployment expenses, ensuring efficient real-time grasp detection.

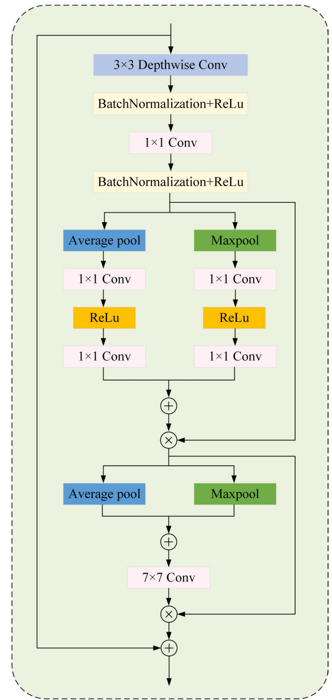

Fig. 2. Structure of VARMC module.

The second section pertains to visual attention mechanisms, enabling the network to shift its focus from global features to the distinctive characteristics of the grasped object while suppressing irrelevant features in the background. Adaptive feature learning is conducted by the network from feature maps computed in two distinct dimensions: channel and spatial. This is aimed at enhancing network accuracy. First, in the channel dimension, global max-pooling and global average-pooling operations are applied separately to the extracted features, resulting in two 1-D feature vectors. Weighted feature allocation is achieved through 1 × 1 convolution, thereby reinforcing the feature information of the grasped object in the channel domain. Second, in the spatial dimension, features extracted from the channel domain are subjected to subsequent max-pooling and average-pooling, compressing them into 2-D feature maps. Subsequently, feature weight assignment is carried out using a 7 × 7 convolution to enhance the spatial-domain feature information.

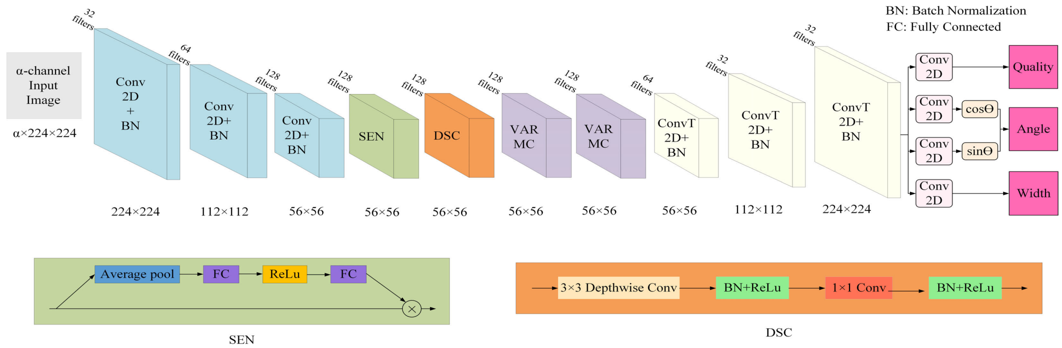

Fig. 3. Lightweight and efficient network architecture GRADSCN for robots to grasp multiple objects in various grasping scenarios.

However, when the network is stacked to a certain depth, it becomes increasingly challenging to train. In addition, a phenomenon of convergence degeneration occurs during the initial stages of network training, leading to a rapid saturation of accuracy while the error rate remains unacceptably high. To address this challenge, we use residual layers, incorporating skip connections that enable direct passage of input signals to subsequent layers while undergoing specific transformations to retain the original input information. Consequently, even if the network is deeper, the network can delve deeper into the acquisition of intricate features, thereby enhancing performance and expressive capacity of the network.

E. Model Structure

Fig. 3 shows our proposed model architecture: GARDSCN. Our network can be trained using single-mode data (RGB or Depth) and is compatible with multimode data (RGBD), which greatly improves the accuracy of the network. The α-channel images serve as the input to the network and undergo feature extraction through three convolutional layers. Batch normalization and ReLu activation functions are applied after each convolutional layer to enhance the representational and generalization capabilities of model, resulting in an output size of $56 \times 56 \times 128$. However, the method of extracting image features by convolutional layer does not clearly model the relationship between feature channels, resulting in some channels contributing less to the grasping task. Therefore, after the three convolutional layers, a squeeze-and-excitation (SEN) module is introduced to direct the focus of network toward salient feature channels. SEN module adaptively learns the importance of each channel and proportionally adjusts the contributions of channels within the feature maps. The SEN module performs average-pooling on the image features extracted by the three convolutional layers. It subsequently uses two fully connected layers to perform dimensionality reduction and then expansion of the feature maps. The first fully connected layer reduces the feature dimensions to the original c/r, followed by the application of a ReLu activation function to introduce nonlinearity. The second fully connected layer is responsible for restoring the original feature dimensions. This set of weights obtained are weighted to the original feature channels one by one through multiplication, and the features of $56 \times 56 \times 128$ are output. To reduce computational complexity and the number of network parameters, depthwise separable convolution (DSC) is introduced after SEN. DSC consists of 3×3 depthwise convolution and 1×1 convolution. The 3×3 depthwise convolution is applied to each channel of the input, enabling the extraction of spatial features on a per-channel basis. The introduced 1×1 convolution facilitates cross-channel feature interactions, thereby merging information from different channels.

To make the network focus on the important grasping area, the VARMC module is introduced. The module integrates channel and spatial dimension information effectively to capture the features of target objects comprehensively. It accelerates computation speed, enhances the saliency of the grasped target within the scene, and effectively mitigates interference from background information. Following the series of convolution layers, the SEN module, the DSC module, and the VARMC module, the size of the images is reduced to $56 \times 56 \times 128$, rendering it challenging for interpretation. Therefore, to facilitate the interpretability of processed images and preserve their spatial features, we use convolutional transpose operations to upsample the images.

The output of GARDSCN contains three parts: grasp confidence Q, grasp angle θ_k, and grasp width W_k. The grasp confidence is a scalar of [0, 1]. The closer the scalar is to 1, the higher the success rate of grasp is. The grasp angle is obtained by $\theta_k = (1/2)\tanh^{-1}((\sin\theta/\cos\theta))$. The grasp width is in pixels. Finally, the pixel grasp point with the highest grasp confidence is selected from the 1-D feature map of grasp confidence, and the grasp angle and grasp width corresponding to the pixel grasp point are selected from the feature map of grasp angle and grasp width to obtain a set of optimal grasping configurations.

Our network comprises 487 080 parameters, making it smaller in scale and faster in computation compared with other networks. This lightweight model structure can be deployed on a physical robot and quickly perform inference grasping to achieve real-time detection and grasping.

F. Training Methods

If a dataset has a sample $E = \{E_1, \ldots E_\phi\}$, the input scene image $F = \{F_1, \ldots F_\phi\}$ and the successful grasp in the image $\gamma_i = \{\gamma_1^1, \ldots, \gamma_{\delta_1}^1, \gamma_1^2, \ldots, \gamma_{\delta_\phi}{}^\phi\}$. In this article, we use the end-to-end model to learn a mapping function

$\beta(F, E) = \gamma_i$. For a successful grasp, the grasp confidence of γ_i is $Q_i \in [0, 1]$. The larger the Q_i, the smaller the negative log-likelihood function corresponding to γ_i. Therefore, the training objective should be a mapping function that minimizes the negative log-likelihood function of the network γ_i

$$\gamma_i = -\frac{1}{\varrho}\sum_{i=1}^{\phi}\frac{1}{\delta_i}\sum_{m=1}^{\delta_i}\log\beta(x_i^b \mid F^i) \tag{6}$$

where ϱ is the number of samples and δ_i is the number of successful grasp for each sample.

The GARDSCN is trained using the Adam optimizer [32] without weight decay or momentum. The learning rate is set to 10^{-3}, with a batch size of 32, and a total of 200 epochs is set. The PyTorch deep learning framework is used to train GARDSCN.

G. Loss Function

By analyzing the operation results of different loss functions under the network, it is found that the smooth L_1 loss can effectively deal with gradient explosion and outlier interference and has strong stability in the training process. Hence, the loss function can be defined as

$$\Upsilon(Z_\alpha, \hat{Z}_a) = \frac{1}{\varrho}\sum_{0}^{\varrho}\Upsilon_\alpha \tag{7}$$

where Υ_α is expressed as

$$\Upsilon_\alpha = \begin{cases} 0.5(Z_\alpha - \hat{Z}_\alpha)^2, & \text{if } \left|Z_\alpha - \hat{Z}_\alpha\right| < 1 \\ \left|Z_\alpha - \hat{Z}_\alpha\right| - 0.5, & \text{otherwise} \end{cases} \tag{8}$$

where α represents the αth grasping; $\hat{Z}_\alpha$ is the grasp pose generated by the GARDSCN; and Z_α is the ground-truth grasp pose.

IV. Experimental Setting

The model proposed in this article necessitates a series of training and evaluation procedures on various publicly available grasp datasets. This section provides an overview of common grasping datasets for evaluation, performance metrics for assessing models, and relevant experimental setups involving robotic systems.

A. Dataset

The number of existing public grasping datasets is already considerable, as shown in Table II. We use Cornell [7] and Jacquard datasets [33] to train our model for evaluation.

1) Cornell Dataset: This dataset contains 885 RGBD images, covering 240 different types of objects. The pixels of each picture are 640 × 480, all of which are single-object scenes. The dataset is manually captured and labeled in the real world. Objects are randomly placed on the experimental platform, and RGBD images are captured using a depth camera. In this work, since our network input is a 224 × 224 pixel picture, we preprocess the dataset to adapt to the network input.

TABLE II
Summary of Existing Public Grasping Datasets

Dataset	Modality	Images	Source	Grasps
Cornell	RGB-D	885	real-world	8019
Jacquard	RGB-D	54K	simulated environment	1.1M
Clutter	RGB	505	real-world	80
Dex-Net 2.0	Depth	6.7M	simulated environment	6.7M
Dex-Net 3.0	Depth	2.8M	simulated environment	2.8M
Dex-Net 4.0	Depth	5M	simulated environment	5M
GraspNet	RGB-D	97k	real-world	3-9M

2) Jacquard Dataset: This dataset consists of 11 619 different scenes from 54 485 different objects, with a total of 4 967 454 crawl annotations. The Jacquard dataset uses a simulated depth camera to capture RGBD images through a pybullet simulated environment and uses a simulated robot to randomly grasp objects multiple times. If the object is successfully grasped, the grasp pose is recorded as a label.

According to prior research [23], [31], [34], we partition the dataset into two subsets, namely, the training dataset and test dataset, using a random approach. The division ratio between the training dataset and test dataset is 9:1.

B. Physical Setup

To perform the grasping experiment on a physical robot, we use the robot Kinova Gen2 with a KG-3 gripper at the end to perform real-world grasping. The grasping experimental device is shown in Fig. 4. In each experiment, objects are randomly placed on the workbench. The robot has 7 DoFs and a working radius of up to 985 mm. With a self-weight of 4.4 kg, it is equipped with a three-finger bionic gripper at the end. The joint is equipped with a torque sensor, and the controller is built in the base of the entire robot. Different from the usual three-finger robot, the three fingers of our robot are distributed on both sides of the palm and can only be closed to the center of the palm. To make the robot perceive the information of the external environment and the target object, we configure a Realsense D435i Depth camera for the robot, which is fixed on the wrist of the robot. The model we propose runs on an Ubuntu 18.04 PC with a 2.5-GHz 13th Gen Intel Core i5-13400F CPU and NVIDIA GeForce RTX 3060 GPU.

C. Grasp Evaluation Metric

For a successful grasp, there are many metrics to evaluate the performance of the predicted grasp pose, including grasp success rate, grasp accuracy, robust grasping rate, and planning time. To reasonably evaluate the experimental results, the rectangular metric proposed by Jiang et al. [7] and the work of Morrison et al. [35] are referred to.

The generation of the grasping results begins by identifying the maximum pixel s* in the grasp confidence map Q to determine the optimal position coordinates of the robot gripper's center in the image space. Following this, the corresponding θ_k and W_k are calculated, which constitute the final grasping rectangle frame for the robot. Ultimately, the predicted

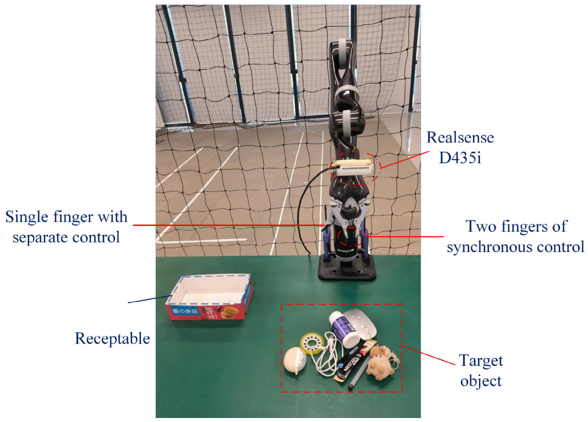

Fig. 4. Robot grasping experimental setup.

grasping rectangle frame is compared with the actual values on the ground. When the output grasp rectangle satisfies the following two conditions, the generated grasping selection is considered to be effective.

1) The intersection over union (IOU) score, denoting the overlap between the actual grasp rectangle and the predicted grasp rectangle at ground level, exceeds 25%.

2) The angular offset between the actual ground grasp rectangle and the predicted grasp direction is less than 30°.

Based on whether the objects in the test dataset images are identical to those in the training dataset images, two methods are used for partitioning the training and test datasets.

1) Imagewise: The objects in the test set all have occurrences in the training dataset, serving as a means to assess the generalization capacity of the GARDSCN across different positions and orientations of the same object.

2) Objectwise: All the objects in the test dataset do not have any occurrences in the training dataset, aimed at evaluating the generalization capability of the GARDSCN with respect to novel objects.

V. Performance on the Public Datasets

In this section, we perform experiments on the publicly available Cornell and Jacquard datasets to validate our proposed model. Because our network is compatible with α-channel data, we test and evaluate different data modes separately. The results show that our model achieves excellent grasp accuracy, fast detection speed, and strong robustness on the Cornell and Jacquard datasets, all of which are higher than the existing grasp detection methods.

A. Grasp Detection on the Cornell Dataset

The dataset is divided into two criteria: imagewise (IW) and objectwise (OW). The IW partitioning method mainly evaluates the ability of the model to detect the grasp positions and poses of objects in different positions and poses, and OW pays more attention to the adaptability of neural network models under novel objects. Since our network is compatible with α-channel images, we perform forward propagation using RGB, Depth, and RGBD as inputs to the network, respectively. The result is shown in Table III. Using RGBD data as input to the network, the accuracy is 98.88% in the IW mode and 96.63% in the OW mode. In addition, our network parameter is only 487 080, and the detection speed of each image is only 28 ms, which can meet the real-time requirements and is suitable for closed-loop grasp. Fig. 5 shows the visualization of our proposed model on the Cornell dataset. The first line indicates the visual grasp with the highest confidence. The last three lines show the corresponding grasp confidence map, grasp angle map, and grasp width map, respectively.

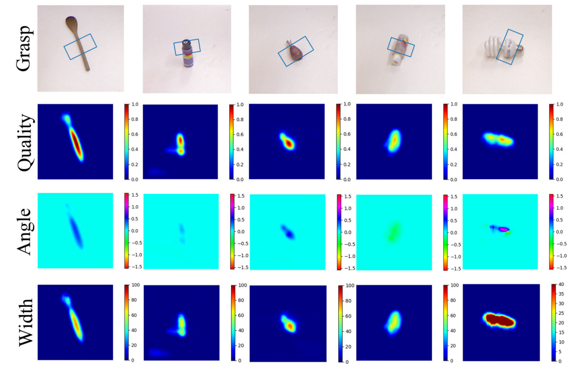

Fig. 5. Experimental results on the Cornell dataset.

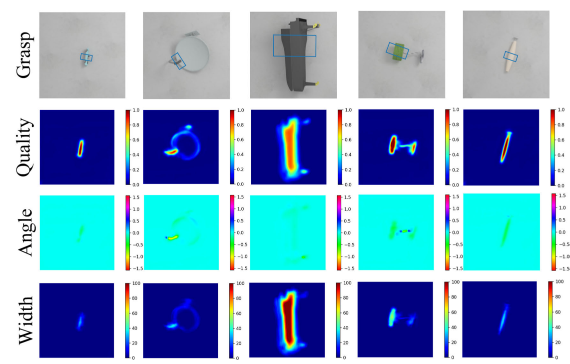

Fig. 6. Experimental results on the Jacquard dataset.

B. Grasp Detection on the Jacquard Dataset

On the Jacqurad dataset, 90% of the data are used to train GARDSCN and 10% to test. We also use single-mode (RGB, Depth) and multimode data (RGBD) for experiments, using RGBD data as input, and we obtain the grasp accuracy of 95.23%, with excellent performance. When using RGB data and Depth data as inputs, we obtain grasp accuracy of 92.62% and 94.18%, which is advanced at the time. Fig. 6 shows the visualization results of our proposed model on the Jacquard dataset. The width of the grasp rectangular autonomously adjusts in response to the size of the object to meet the criteria for stable grasping conditions. It should also be noted that even when the color of the object is similar or the same as the background color, our model still accurately outputs the grasp rectangle due to its attention to channel information, as shown in Fig. 7. Due to the large amount of data in the Jacquard dataset, it is sufficient to prove that GARDSCN has the ability to grasp any type of object.

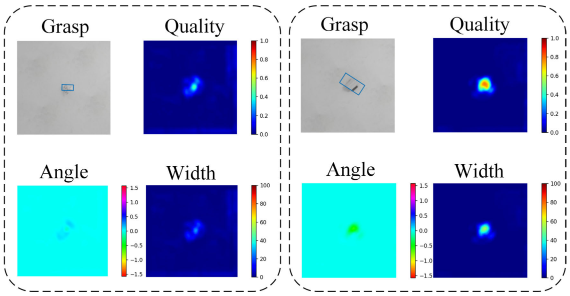

Fig. 7. Grasp detection when the color of objects is similar or the same as the background.

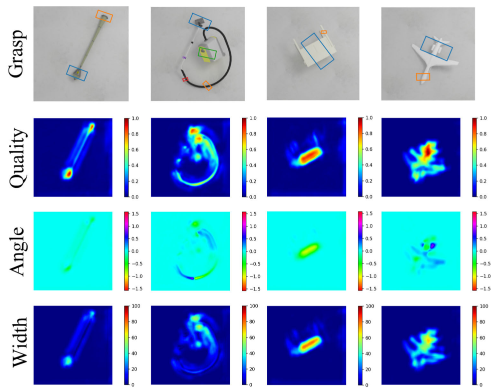

Fig. 8. Single-object multiple grasp.

Nonetheless, considering the complexity of the grasping environment and the properties of objects, predicting only a single grasping representation model may be insufficient for the grasping task. Specifically, the grasping points predicted on both sides of the object may result in slippage during the grasping process due to smooth surfaces. Therefore, a strategy of multiple grasping poses is proposed for a single object. Fig. 8 shows multiple grasps on a single object. The single-object multigrasp strategy provides multiple feasible and high-quality grasp poses for a given object.

C. Comparative Experiments

1) Cornell Dataset: On the Cornell dataset, our method is compared with the existing methods of GRCNN and GGCNN, and the results of grasping detection with the highest grasp confidence are, respectively, given. Fig. 9 indicates the detection results of the unseen novel objects captured by GARDSCN, GRCNN, and GGCNN. It can be seen that GGCNN and GRCNN cannot well-distinguish the area where the target object exists, and the best grasp position detected is not really the optimal grasp position, resulting in the robot being unable to grasp the object stably. However, our proposed model GARDSCN does not have such problems. In contrast, GARDSCN has excellent grasp quality.

In the real-time detection of robot grasping, accuracy and rapidity are undoubtedly two important factors to measure the quality of the network model: accuracy determines the positioning ability of the robot grasping, and rapidity determines whether the designed model can carry out real-time operation of the robot grasping detection. Table III shows comparison of the performance of GARDSCN with some existing advanced methods on the Cornell dataset. When using RGBD data input, the grasp accuracy of GARDSCN in IW segmentation and OW segmentation is 98.88% and 96.63%, respectively, which is much higher than other methods. In terms of inference speed, the GARDSCN can infer the grasping pose of each object in just 28 ms. Although the inference speeds of Wu et al. [22], Park and Chun [38], and Wang et al. [39] are slightly superior to GARDSCN, their detection accuracies are much lower than that of GARDSCN. Our method strikes a good balance between grasp accuracy and inference speed.

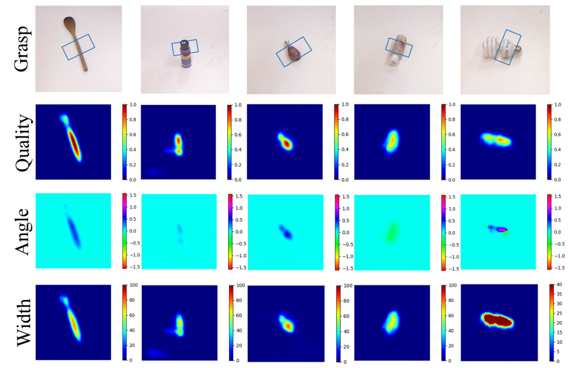

Fig. 9. Comparison studies on the Cornell dataset.

TABLE III

Performance of Different Approaches on the Cornell Dataset

Author	Approach	Input Modality	Accuracy(%)		Speed(ms)
			IW	OW	
Jiang [7]	Fast Search	RGB-D	60.5	58.3	5000
Lenz [18]	SAE, struct. reg.	RGB-D	73.9	75.6	1350
Yu [36]	Multilevel CNNS	RGB	95.8	96.2	-
Morrison [35]	GGCNN	D	73.0	69.0	19
Asif [37]	Graspnet	RGB-D	90.2	90.6	24
Wu [22]	PLG-Net	D	92.6	-	6.8
Guo [20]	ZF-net	RGB-D	93.2	89.1	-
Park [38]	STN	RGB-D	89.6	-	23
Wang [39]	Fully convolution neural network	RGB-D	94.42	91.02	8
Wang [40]	SGDN	RGB	96.8	92.27	51.6
Ours	GARDSCN	RGB	97.75	94.38	**28**
		D	94.38	92.13	
		RGB-D	**98.88**	**96.63**	

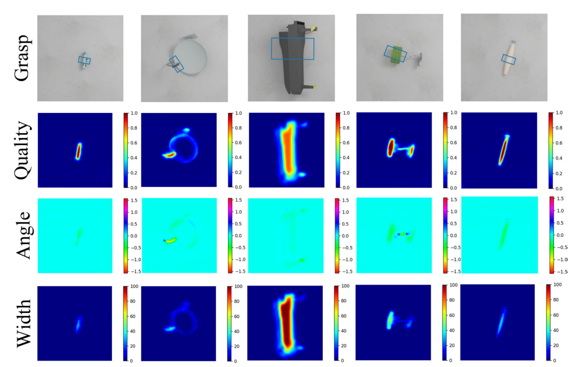

Fig. 10. Comparison studies on the Jacquard dataset.

2) Jacquard Dataset: As shown in Table IV, our method is compared with some state-of-the-art approaches in recent years on the Jacquard dataset. Fig. 10 illustrates the detection results of different methods on unseen objects on the Jacquard dataset. It can be readily observed from Fig. 10.

TABLE IV
PERFORMANCE OF DIFFERENT APPROACHES ON THE JACQUARD DATASET

Author	Algorithm	Input Modality	Accuracy(%)
Morrison [41]	GGCNN2	D	84.0
Zhou [42]	FCGN, ResNet-101	RGD	91.8
Song [43]	Single-stage Network	RG-D	93.2
Zhang [44]	MSROI-GD	RGB	90.4
Liu [27]	Y-Net	RGB-D	92.1
Ours	GARDSCN	RGB	92.62
		D	94.18
		RGB-D	**95.23**

TABLE V
GRASP ACCURACY OF DIFFERENT APPROACHES BASED ON DIFFERENT IOU STANDARDS ON THE CORNELL DATASET

Approach	IOU threshold				
	0.25	0.30	0.35	0.40	0.45
Morrison [41]	87.6	84.2	82.5	79.7	72.3
Chu [21]	96.0	94.9	92.1	84.7	-
Zhou [42]	98.31	97.74	96.61	95.48	-
Kumra [45]	92.1	89.9	88.8	84.3	83.1
Ours	**98.88**	**98.88**	**96.63**	**93.26**	**87.64**

1) For the grasp of toy model beds, GRCNN and GGCNN exhibit a highly erroneous selection of grasp positions, resulting in a grasp that fails to enable the robot to securely manipulate the object.
2) For the grasp of the target, although the grasp position chosen by GRCNN and GGCNN seems to be correct, the grasp angle is problematic. The grasp of these two methods will cause the end gripper of the robot to contact and collide with the three extensions on the right side or cause the object to deform during the real grasp, which is a dangerous grasp behavior. Furthermore, the grasp width of GGCNN is incorrect.

In contrast, due to the incorporation of our designed module of VARMC within our network, it enables the enhancement of pertinent features while suppressing less consequential ones. More importantly, it addresses the issue of gradient vanishing that arises with the increase in the number of network layers. Therefore, the model GARDSCN can capture the details of the target object well and improve the grasp accuracy. Moreover, the grasp rectangle generated by GARDSCN tends to the middle position of the object and is more accurate, which improves the stability of the robot to grasp the target object.

3) Stability Analysis: Stability is an important index in robot grasping detection, which determines whether the robot can be robust in complex dynamic environments. The performance of GARDSCN is evaluated using the criteria of different IOU and compared with other network models on the Cornell dataset, including those designed by Morrison et al. [41], Chu et al. [21], Zhou et al. [42], and Kumra et al. [45]. The accuracy of various methods under different IOU standards is shown in Table V. The proposed network GARDSCN exhibits the highest accuracy under different IOU standards. As the IOU standard increases, the accuracy of GARDSCN remains consistently stable. Even at an IOU of 0.45, the accuracy of the network reaches 87.64%. This indicates that the network designed in this article demonstrates increased stability, enhanced robustness, and a competitive performance.

4) Results Analysis: Some of the current state-of-the-art methods have the following limitations.

1) When the color of the target object is the same or similar to the color of the workbench background, these methods cannot accurately distinguish between the target object and the workbench background.
2) In the face of some complex objects, the grasp rectangle cannot accurately give the optimal grasp pose. In the area that is easy to grasp, the network model gives a very low grasp quality score.
3) Current works all design networks under ideal constraints, making them unable to adapt to complex and variable grasping environments.

The good news is that we introduce the GARDSCN, which supports the input of α-channel data, significantly enhancing the grasping success rate of robots in diverse scenarios. The designed VARMC module in the network focuses on both channel and spatial dimensional information, adaptively elevating the weights of critical features of target objects and suppressing the interference from the grasping background. More importantly, our proposed systematic robot grasping system integrates grasping detection with grasping execution planning, enabling efficient completion of grasping tasks.

VI. EXPERIMENTS

In addition to validating the model on two public datasets, a series of grasping experiments are also conducted in the real world using the Kinova Gen2 robot. To better simulate the complex and variable environments in the real world, various experimental settings are established, including multicolored grasping backgrounds, background distractions, and changes in lighting conditions. Furthermore, given the existence of diverse grasping objects in real-world scenarios, we select flexible dolls, water bottles, snacks, and common household items as unknown grasping objects and place them in different grasping environments for experiments. The results demonstrate that GARDSCN achieves high grasping accuracy while maintaining fast grasping pose inference, exhibiting strong robustness against background distractions and lighting variations. Our method integrates the ROS interface, enabling it to be mounted on any robot for grasping tasks, and can be applied in complex and variable scenarios such as warehousing logistics, home services, and industrial manufacturing.

A. Grasping in Single-Object Scene

Our Kinova Gen2 is a robot with a three-finger gripper at the end, in which two symmetrical fingers can continue to move in sync (simultaneously contracting inward or opening outward side), and the other third finger can be controlled separately. To facilitate the deployment of the proposed model and the consistency comparison with the current algorithm, all

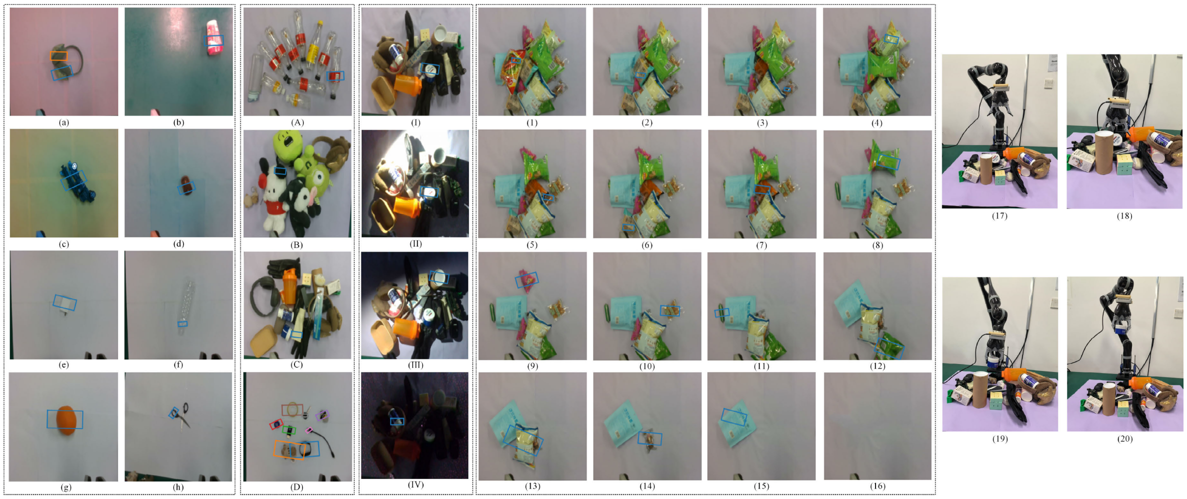

Fig. 11. Conducting grasping experiments in the real world using the Kinova Gen2 robot. (a)–(h) Detection of grasping in different scenarios of single-object grasping. (A)–(D) Grasping in cluttered scenes with multiple objects of different types. (I)–(IV) Grasping objects in cluttered scenes under varying lighting conditions. (1)–(16) Sequential grasp pose inference for objects in a cluttered scene. (17) and (18) Visual perception module obtains the external scene information and target object information, and the detection and inference module calculates the grasp pose and selects the grasp pose with the highest grasp quality score. (19) Kinova Gen2 robot securely grasps the target object with a high-quality grasp pose and slowly lifts it upward. (20) Grasp planning module designs a safe target path and controls the robot to transport the object to the designated location.

the grasping experiments only use two symmetric fingers for grasping.

In the single-object scenarios, objects with varying shapes, colors, sizes, and materials are randomly placed in different positions and orientations. As shown in Fig. 11(a)–(h), our robotic system uses the Realsense D435i camera to capture target information, followed by generating the highest quality grasp pose representation through the GARDSCN in the detection and inference module of the robotic grasping system. Fig 11(a)–(d) demonstrates grasping experiments conducted in distinct environments, each with a unique background color. Fig. 11(a) demonstrates the grasp pose representation model for an earmuff, with two representations indicating equivalent grasp quality. In Fig. 11(b), despite background interference and light reflection, GARDSCN accurately locates the target object and outputs the grasp representation model. Fig. 11(c) presents the grasp representation for an object with complex surface texture information. In Fig. 11(d), the grasp representation is outputted amidst uneven background color distribution. In addition, we conducted experiments in four different scenarios, including those with similar target and background colors, transparent objects, spherical objects, and objects with complex edge information, as shown in Fig. 11(e)–(h). In scenarios where object colors are extremely similar to background colors and background interference is present, this work addresses an aspect overlooked by previous researchers. Our model maintains the ability to accurately locate objects and output highly confident grasps, demonstrating robustness against diverse backgrounds colors and interferences. In scenes with transparent objects, the model similarly outputs accurately positioned and posture-compliant grasp poses that align with real-world grasping conditions, overcoming the current limitations of depth cameras in accurately measuring transparent objects. In scenarios involving spherical objects and objects with complex edge information, GARDSCN adaptively adjusts the grasp width and angle to achieve high-quality grasping.

TABLE VI
COMPARISON OF EXPERIMENTAL RESULTS OF PHYSICAL ROBOT GRASPING

Approach	Robot platform	Grasp success rate (%)	
		Single-object	Clutter
Dex-Net2.0 [46]	ABB YuMi	80	-
GGCNN2 [41]	Kinova Mico	92.0	87.0
GRCNN [45]	Baxter	95.4	93.5
DSQNet [17]	Franka Emika	93.0	-
GSMR-CNN [24]	Franka Emika	-	76.4
Anygrasp [13]	UR5	-	93.3
Ours	Kinova Gen2	**96.67 (290/300)**	**94.1**

Thirty similar objects are selected for single-object scenario experiments, with each object tested in ten different positions and poses. Out of a total of 300 grasp attempts, 290 are successful, resulting in an accuracy rate of 96.67%. The grasping results of our physical robot, the Kinova Gen2, are compared with those obtained on different robots by other researchers, as shown in Table VI.

B. Grasping in Multiobject Cluttered Scene

In application scenarios such as express parcel sorting, assembly line production, and warehousing, where a multitude of items are involved, their placement may not necessarily adhere to a structured arrangement, often leading to instances of stacking. Therefore, it is crucial to conduct grasping experiments in cluttered scenes with multiple objects. We assemble a set of grasping objects from previously unseen items and randomly place them on a workspace to simulate these cluttered scenarios. Fifty cluttered scenes are created, each containing 8–20 objects, encompassing both multiple objects of the same category and those of different categories. As shown in Fig. 11(A)–(C), representative cluttered scenes include sets of cups and empty bottles, flexible dolls, and household items. GARDSCN can adapt to the uncertainty of these grasping objects and demonstrate excellent grasping performance. Fig. 11(D) depicts the grasp prediction of GARDSCN for a set of small-sized grasping objects in a cluttered scene. Distinguished from prior works [35], [41], [47], our GARDSCN can output very high-quality grasp rectangles for multiple objects at one time in complex multiobject scenarios.

Furthermore, a series of grasping experiments are conducted under various lighting conditions, ranging from bright, moderately bright, dim, to dark conditions, to infer grasp poses for multiple objects in cluttered scenes. The results indicate that our proposed method maintains stable grasping performance even under different lighting conditions, including extreme darkness. In Fig. 11(1)–(16), we visualize the grasp pose representation models for deformable flexible packaged snacks in a cluttered scene. Specifically, the highest grasp pose model for each object is predicted, it is grasped and transported to the target location using the robot, and then the process for the next object is repeated until no objects are visible to the camera. Fig. 11(17)–(20) shows the Kinova Gen2 robot in the process of grasping objects.

Ultimately, an average grasping success rate of 94.1% is achieved across 50 cluttered scenes. When compared with recent typical methods, our approach achieves the highest grasping success rate, as shown in Table VI. Our robotic intelligent grasping system exhibits grasping capabilities for arbitrary scenes and objects, generating high-quality grasping poses for previously unseen objects at millisecond speeds, and adapting to various uncertainties. It can perform safe and efficient grasping tasks even under poor lighting conditions, demonstrating potential in industrial manufacturing, household services, and express delivery.

C. Failure Cases Analysis

In the few cases where we fail, the following reasons exist in addition to the network GARDSCN itself giving a low-confidence grasp as follows.

1) The object has a smooth surface, which leads to the lack of sufficient friction when the finger is closing, resulting in the object slipping during the grasp process.
2) Errors generated during hand–eye calibration tasks. Errors during the hand–eye calibration task ultimately affect the robot body to confirm the position of the target object, even if the RGBD camera has precisely positioned the target.

VII. Conclusion

In this article, we propose a robotic intelligent grasping system integrating visual perception, detection, and inference and grasping planning. The designed GARDSCN resolves the grasping challenges of multiple objects in diverse scenarios. It achieves remarkable grasping accuracy on two datasets: Cornell and Jacquard. In real-world complex and diverse scenarios, the Kinova Gen2 robot can generate grasping poses for arbitrary unseen objects at millisecond speeds, insensitive to backgrounds and lighting conditions. Our method enables high-speed grasping of arbitrary objects in arbitrary scenes, showing potential applications in express delivery, warehousing, and logistics. In future work, we plan to design visual servo control algorithms to achieve stable grasping during dynamic processes.

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Hejia Gao (Member, IEEE) received the B.Eng. degree in intelligence science and technology and the Ph.D. degree in control science and engineering from the School of Automation and Electrical Engineering, University of Science and Technology Beijing, Beijing, China, in 2016 and 2021, respectively.

She was a Visiting Researcher at the Department of Mechanical, Industrial and Aerospace Engineering, Concordia University, Montreal, QC, Canada, for 15 months, in 2019. She is currently an Associate Professor with the School of Artificial Intelligence, Anhui University, Hefei, China. She has published more than ten international journal articles, and two articles have been ranked as ESI Highly Cited Paper. Her research interests include neural networks, reinforcement learning, flexible systems, and vibration control.

Dr. Gao is the Co-Chair of the IEEE 37st Youth Academic Annual Conference of Chinese Association of Automation. She has been an Associate Editor of the 2019 IEEE International Conference on Advanced Robotics and Mechatronics. She is currently serving as the Guest Editor of *Advances in Mechanical Engineering*.

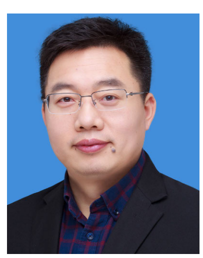
Junjie Zhao received the B.Eng. degree in automation from Nanjing University of Information Science and Technology, Nanjing, China, in 2022. He is currently pursuing the M.S. degree with Anhui University, Hefei, China.

Mr. Zhao once received the Bronze Prize of the 4th China College Students' "Internet Plus" Innovation and Entrepreneurship Competition and participated in an Innovation and Entrepreneurship Training Program for college students in Jiangsu Province and has completed the project.

Juqi Hu (Member, IEEE) received the M.S. degree in mechanical and electronic engineering from the School of Mechatronic Engineering and Automation, Shanghai University, Shanghai, China, in 2014, and the Ph.D. degree in mechanical engineering from the Department of Mechanical, Industrial and Aerospace Engineering, Concordia University, Montreal, Quebec, Canada, in 2021.

He was a group member of Networked Autonomous Vehicles (NAV) Laboratory at Concordia University. He is currently an Assistant Professor with the School of Artificial Intelligence, Anhui University, Hefei, China. His research interests include tire-road friction coefficient estimation, trajectory planning, and tracking for autonomous vehicles.

Dr. Hu was a recipient of the final list for the Young Author Award from the 10th IFAC Symposium on Intelligent Autonomous Vehicles (IAVs). He has served as a reviewer for several journals such as the IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS and IEEE TRANSACTIONS ON INTELLIGENT VEHICLES.



Changyin Sun (Senior Member, IEEE) received the B.Eng. degree in applied mathematics from the College of Mathematics, Sichuan University, Chengdu, China, in 1996, and the M.S. and Ph.D. degrees in electrical engineering from Southeast University, Nanjing, China, in 2001 and 2004, respectively.

He is currently a Professor with the School of Automation, Southeast University. He has coauthored four books and published more than 160 international journal articles. His research interests include intelligent control, flight control, pattern recognition, and optimal theory.

Prof. Sun is a Chinese Association of Automation Fellow. He is serving as an Associate Editor of IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, *Neural Processing Letters*, and IEEE/CAA JOURNAL OF AUTOMATICA SINICA.